

PROJECT TERRAPULSE

Multi-Source Market Intelligence & Data Warehouse

Technical Report

SQL Capstone Project

Data Analytics & Database Engineering

Instructor: Peter Pann

Platform: MySQL 8.0 | MySQL Workbench

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Date: [September 22, 2026]

Executive Summary

TerraPulse Global Ltd. required a centralized data warehouse to unify five disconnected data sources — marketplace orders, logistics fulfillment, customer demographics, a supplier product catalog, and foreign-exchange rates — into a single, query-ready analytical platform. This report documents the design, build, and validation of that warehouse in MySQL 8.0, covering six phases: schema design, ingestion, ETL cleaning, business-intelligence querying, transactional integrity testing, and a reusable reporting layer of views.

The finished warehouse holds 200 products, 15,000 customers, and 500,000 orders with matching logistics records, all converted to a single reporting currency (USD) and validated end-to-end: every raw record loaded into staging was successfully transformed and loaded into the production schema with zero record loss.

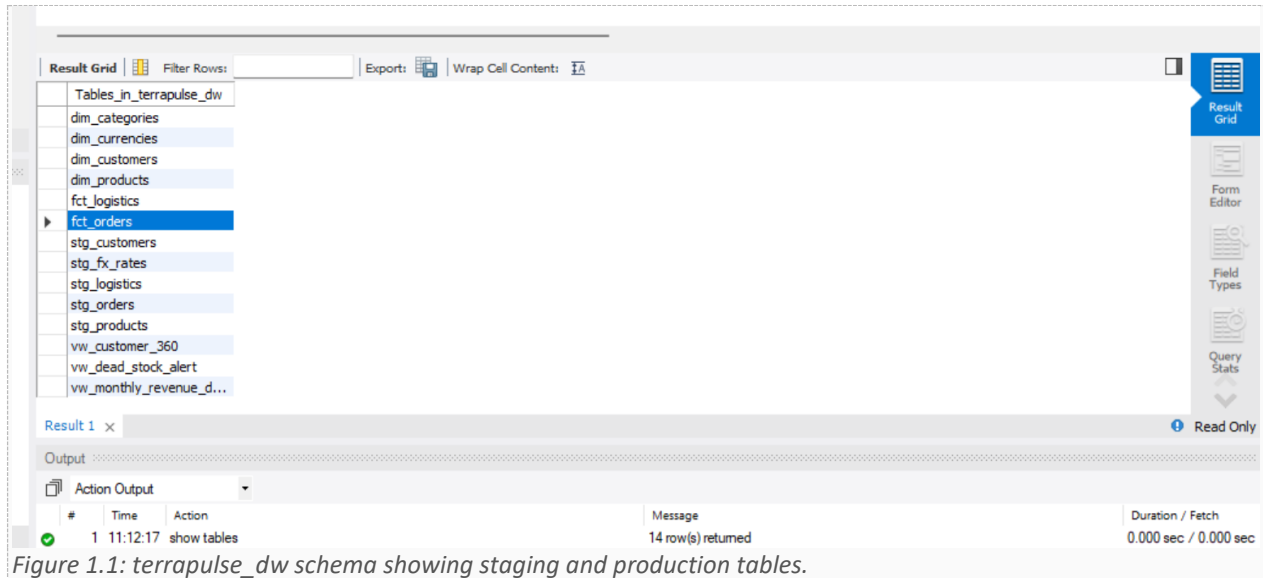
Three reusable views were built for ongoing business reporting (monthly revenue, customer 360, and dead-stock alerting), and a transaction-safety exercise demonstrated recovery from a simulated data-entry error using savepoints, without losing valid work already committed in the same session.

Phase 1: Database Architecture

The warehouse follows a two-layer design. A staging layer (prefixed `stg_`) accepts raw data exactly as supplied by each source, with every column typed as `VARCHAR` and no constraints, so that malformed or inconsistent values do not block ingestion. A production layer (prefixed `dim_` for dimensions and `fct_` for facts) holds fully typed, constrained data: primary keys, foreign keys, `NOT NULL` rules, and sensible defaults. This separation isolates data-quality problems in staging so they can be inspected and fixed before touching the trusted, query-facing schema.

1.1 Schema Overview

Layer	Table	Purpose
Staging	<code>stg_orders</code>	Raw marketplace order records, mixed date/status/currency formats
Staging	<code>stg_logistics</code>	Raw shipment and carrier records
Staging	<code>stg_customers</code>	Raw customer demographic records (JSON source)
Staging	<code>stg_products</code>	Raw supplier product catalog
Staging	<code>stg_fx_rates</code>	Raw currency exchange rates (JSON source)
Production	<code>dim_currencies</code>	Currency code, name, and USD conversion rate
Production	<code>dim_categories</code>	Distinct product categories
Production	<code>dim_products</code>	Typed product master data with cost, price, stock
Production	<code>dim_customers</code>	Typed, standardized customer master data
Production	<code>fct_orders</code>	Typed order transactions with multi-currency conversion
Production	<code>fct_logistics</code>	Typed shipment records linked to orders



1.2 Referential Integrity

fct_orders enforces foreign keys against dim_customers, dim_products, and dim_currencies, so an order cannot be loaded for a customer, product, or currency that does not exist in the production schema. fct_logistics enforces a foreign key against fct_orders, ensuring every shipment record ties back to a real order.

Phase 2: Ingestion & Staging

All five raw sources were loaded into their matching staging tables. CSV and TSV files (orders, logistics, supplier catalog) were loaded with LOAD DATA INFILE; the two JSON sources (customer demographics and FX rates) were parsed using MySQL's JSON_TABLE() function to unpack nested arrays into flat staging rows.

2.1 Ingestion Verification

Row counts were checked immediately after each load to confirm nothing was dropped or truncated during ingestion.

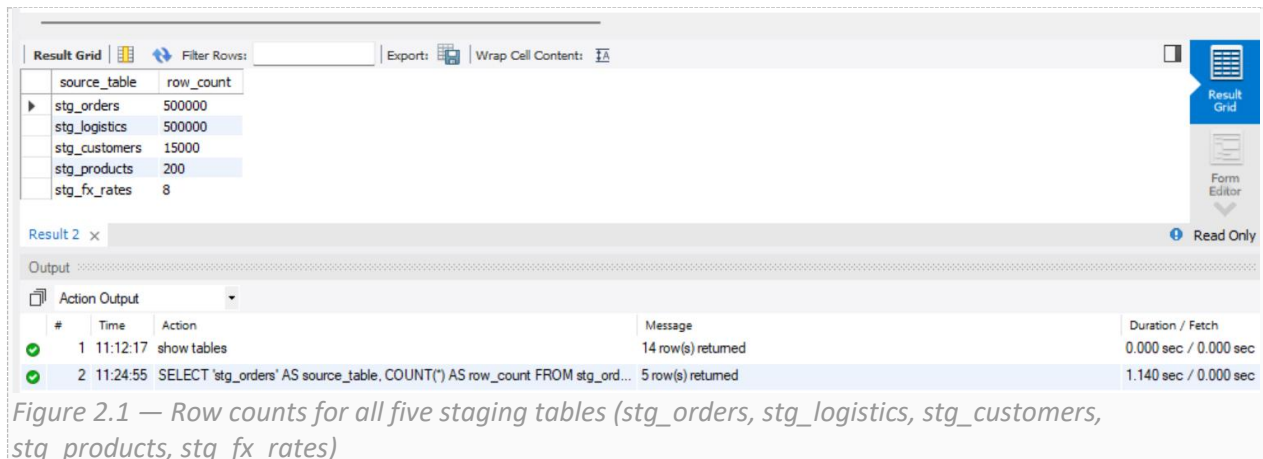


Figure 2.1: Row counts confirming successful ingestion into staging.

2.2 Data Quality Audit

Before any cleaning was applied, a data-quality audit was run against the staging tables to document the scale of the problems the ETL pipeline needed to solve:

- Missing phone numbers and email addresses in stg_customers
- Missing loyalty scores in stg_customers
- Multiple casing/whitespace variants of order status in stg_orders (e.g. "DELIVERED", "delivered", " Delivered ")
- Eight distinct currency codes requiring FX conversion
- Four different date formats across order, registration, and ship dates

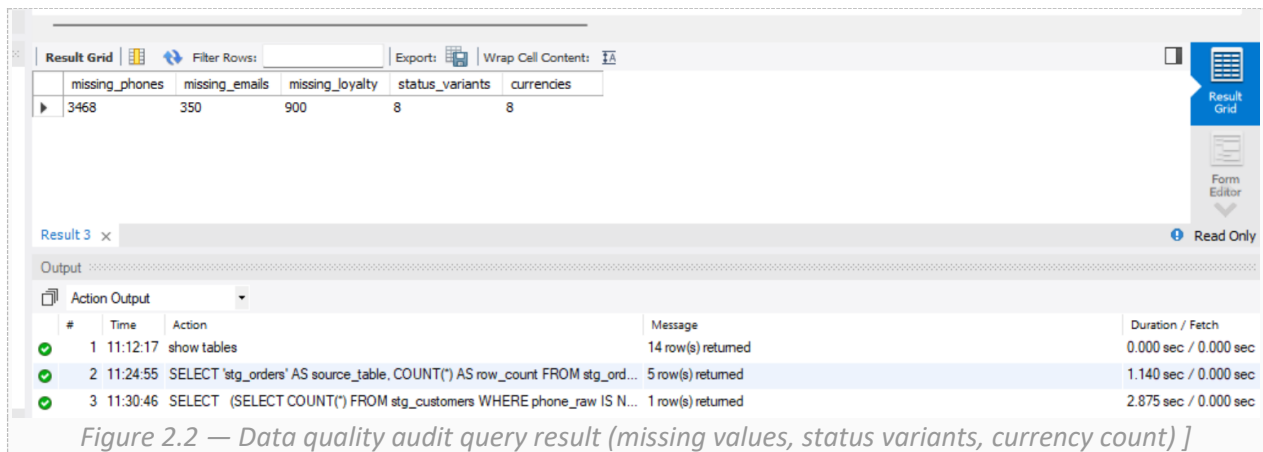


Figure 2.2: Data quality issues identified in staging before cleaning.

Phase 3: ETL Cleaning Pipeline

The cleaning pipeline transforms every staging table into its production counterpart, resolving the issues identified in the data-quality audit.

3.1 Currencies, Categories & Products

- Currency codes and FX rates were cast to their proper numeric types and loaded into dim_currencies.
- Distinct product categories were extracted from the raw catalog into dim_categories.
- Products were loaded into dim_products with cost, retail price, and stock cast to DECIMAL/INTEGER, joined to their category via dim_categories.

3.2 Customer Standardization

- Names were re-cased to proper case (first letter uppercase, remainder lowercase) to resolve inconsistent casing.
- Missing loyalty scores were defaulted to 500; missing country was defaulted to 'Unknown'.

- Registration dates were normalized from four different raw formats (ISO, US, UK, and "DD-Mon-YYYY") into a single DATE type using pattern matching and conditional parsing.

3.3 Order Status & Multi-Currency Conversion

Each order's native-currency unit price was converted to USD using the matching FX rate from dim_currencies. Gross amount, discount amount, shipping fee, and net revenue were all derived in USD, giving a single consistent currency for every downstream report regardless of which of the eight source currencies the order was originally placed in.

3.4 Logistics Cleaning

Carrier names were re-cased and blank values defaulted to 'Unassigned'. Ship dates were normalized using the same multi-format date logic as orders and customers. Shipping lead time was derived as the number of days between order date and ship date.

3.5 Validation: Zero Record Loss

After cleaning, staging and production row counts were compared directly. Every table matched exactly, confirming that no records were silently dropped during type casting, joins, or date parsing.

Table	Raw (staging)	Clean (production)	Match
Products	200	200	✓
Customers	15,000	15,000	✓
Orders	500,000	500,000	✓
Logistics	500,000	500,000	✓

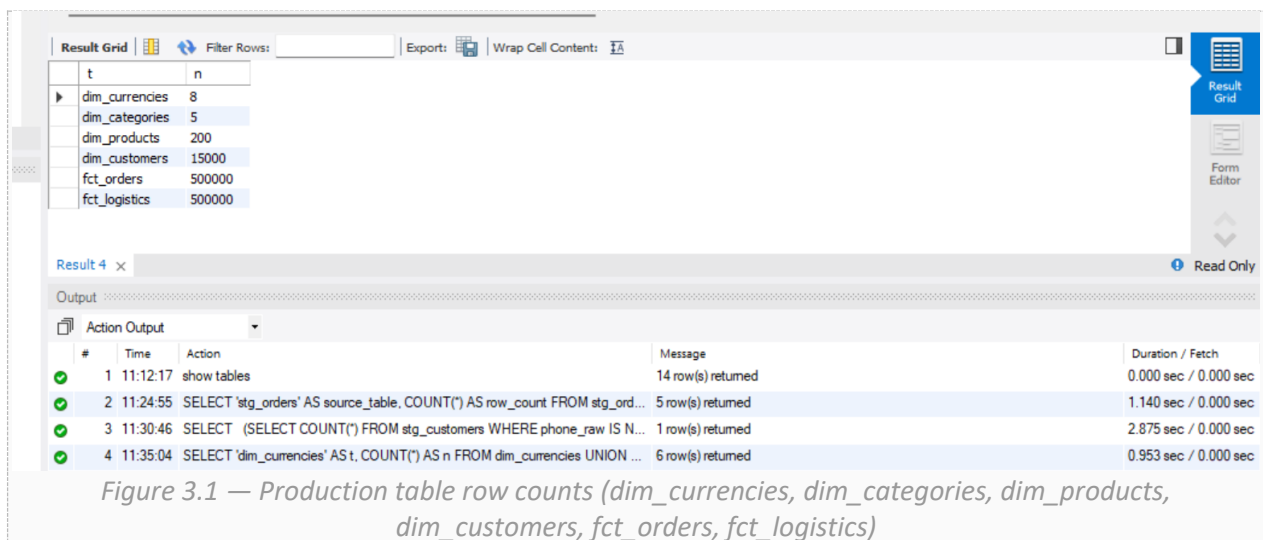


Figure 3.1: Row counts in production tables after ETL cleaning, matching staging exactly.

Phase 4: Business Intelligence Queries

Eight analytical queries were written against the production schema to answer specific business questions. Each is summarized below.

Q4.1 — Revenue & Profitability by Product Category

Aggregates units sold, gross revenue, discounts, net revenue, cost, gross profit, and profit margin percentage by category, restricted to Delivered and Shipped orders.

category_name	units_sold	net_revenue	gross_profit
EV Mobility & Charging	268977	633271177.58	194330743.23
Industrial Energy Efficiency	268518	478034221.51	168144341.85
Renewable Energy & Solar	271263	220413645.42	56794390.77
Smart Grid & IoT Energy	268006	203972500.19	71298587.50
Eco-Friendly Home & Climate	268232	170548321.83	58421060.55

#	Time	Action	Message	Duration / Fetch
3	11:30:46	SELECT (SELECT COUNT(*) FROM stg_customers WHERE phone_raw IS ...	1 row(s) returned	2.875 sec / 0.000 sec
4	11:35:04	SELECT 'dim_currencies' AS t, COUNT(*) AS n FROM dim_currencies UNIO...	6 row(s) returned	0.953 sec / 0.000 sec
5	11:43:39	SELECT customer_id, first_name, last_name, full_name, country, loyalty_scor...	10 row(s) returned	0.000 sec / 0.000 sec
6	11:47:25	SELECT dc.category_name, SUM(fo.quantity) AS units_sold, SUM(fo.net_rev...	5 row(s) returned	6.063 sec / 0.000 sec

Figure 4.1 — Q4.1 result: revenue and profitability by category]

Q4.2 — Geographic Revenue Distribution

Ranks countries by order count and net revenue, and calculates each country's percentage share of global revenue.

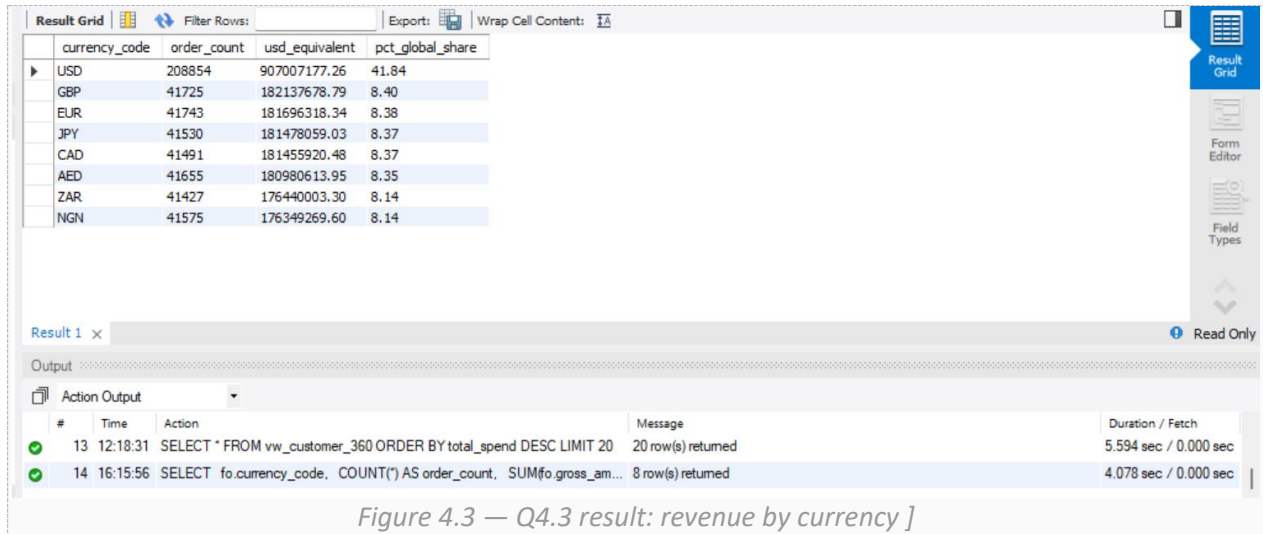
country	orders	net_revenue
Germany	43839	179531096.19
USA	43257	179004107.98
Kenya	42044	176521614.22
Japan	42686	176166524.73
India	42637	175733125.67
Ghana	42076	175342884.08
Nigeria	41877	172382851.20
United Kingdom	41527	170298706.30
South Africa	40867	169879158.43
Canada	40971	168914471.65
UAE	39000	162959454.19
Australia	39219	159993953.98

#	Time	Action	Message	Duration / Fetch
6	11:47:25	SELECT dc.category_name, SUM(fo.quantity) AS units_sold, SUM(fo.net_rev...	5 row(s) returned	6.063 sec / 0.000 sec
7	11:49:01	SELECT dcu.country, COUNT(DISTINCT fo.order_id) AS orders, SUM(fo.net...	12 row(s) returned	4.031 sec / 0.000 sec

Figure 4.2 — Q4.2 result: revenue by country]

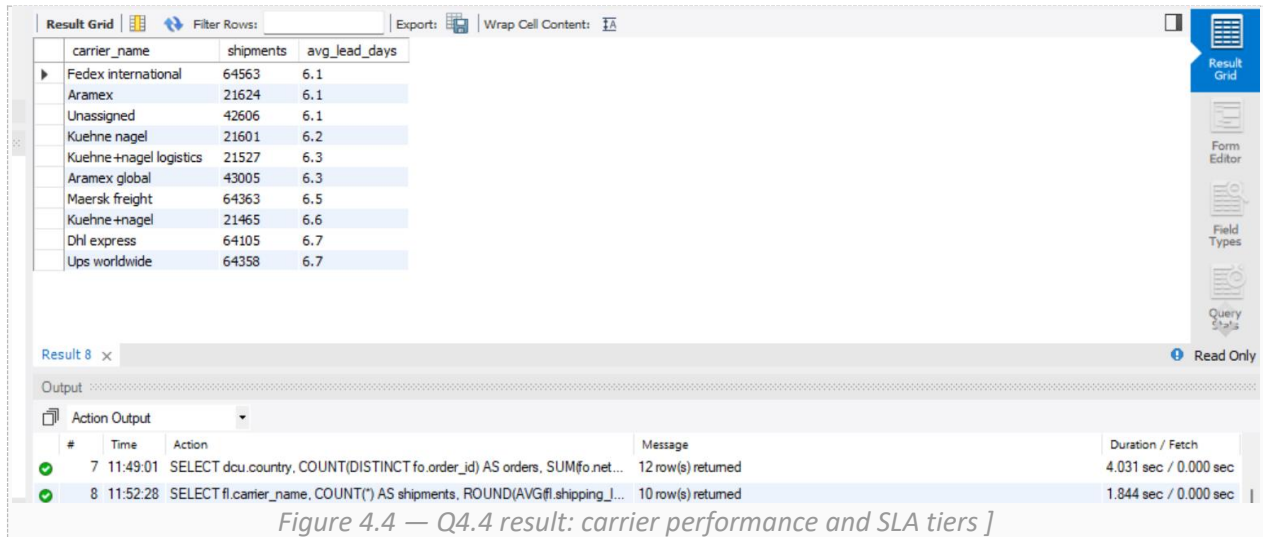
Q4.3 — Currency Exposure Analysis

Quantifies how much revenue flows through each of the eight transaction currencies, identifying the company's largest non-USD exposure.



Q4.4 — Carrier SLA Performance Audit

Ranks logistics carriers by average shipping lead time and classifies each into an SLA tier (Excellent / Acceptable / At Risk).



Q4.5 — Customer Value Segmentation

Segments every customer into Platinum, Gold, Silver, or Bronze based on lifetime spend, ranked by total value.

customer_id	full_name	lifetime_spend	order_count	customer_segment
13100	Lucas Vance	528136.76	34	Platinum
10940	Darius Boateng	521256.74	42	Platinum
10271	Josief Okonkwo	506129.86	34	Platinum
9366	Josief Mensah	487784.55	35	Platinum
14352	Valentina Goldberg	487338.58	43	Platinum
6294	Anna Brown	476445.12	45	Platinum
11669	Josief Brown	464010.09	50	Platinum
7409	Chioma Brown	463636.52	49	Platinum
4093	Darius Sterling	457909.09	37	Platinum
3365	Chioma Appiah	457858.09	47	Platinum
8102	Zainab Okonkwo	457237.37	43	Platinum
15093	Amara	456003.20	35	Platinum
13940	Zainab Dubois	448840.88	34	Platinum
11069	Lucas Vance	443974.28	37	Platinum

#	Time	Action	Message	Duration / Fetch
14	16:15:56	SELECT fo.currency_code, COUNT(*) AS order_count, SUM(fo.gross_am...	8 row(s) returned	4.078 sec / 0.000 sec
15	16:18:03	SELECT dcu.customer_id, dcu.full_name, SUM(fo.net_revenue_usd) AS If...	15 row(s) returned	4.641 sec / 0.000 sec

Figure 4.5 — Q4.5 result: top customers by lifetime spend]

Q4.6 — Dormant Customer Anti-Join

Uses a LEFT JOIN / IS NULL anti-join to identify customers who have registered but never placed an order, ranked by days since registration.

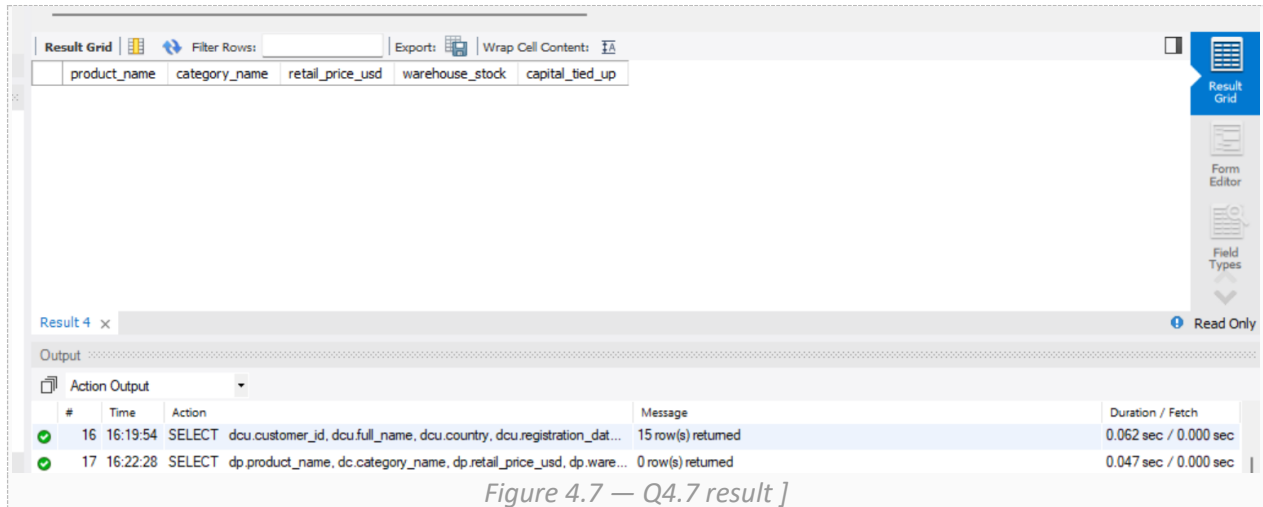
customer_id	full_name	country	registration_date	days_since_registration
14517	Anna Schwarz	Germany	2020-01-01	2456
2546	Sophie Brown	UAE	2020-01-03	2454
1014	Anna Nakamura	Canada	2020-01-06	2451
15790	Sipho Sterling	Australia	2020-01-06	2451
13679	Bruno Silva	United Kingdom	2020-01-08	2449
7377	Amira Nwachukwu	South Africa	2020-01-08	2449
5014	Emeka Nakamura	Ghana	2020-01-09	2448
3314	Emeka Vance	Kenya	2020-01-15	2442
6486	Kevin Dubois	UAE	2020-01-16	2441
7752	Fatima	South Africa	2020-01-18	2439
9904	Klaus	Nigeria	2020-01-18	2439
4904	Abena Mensah	USA	2020-01-24	2433
4179	Josief Osei	Japan	2020-01-25	2432
12986	Kevin Eriksson	Japan	2020-01-26	2431
15160	Fatima Osei	Kenya	2020-01-31	2426

#	Time	Action	Message	Duration / Fetch
15	16:18:03	SELECT dcu.customer_id, dcu.full_name, SUM(fo.net_revenue_usd) AS If...	15 row(s) returned	4.641 sec / 0.000 sec
16	16:19:54	SELECT dcu.customer_id, dcu.full_name, dcu.country, dcu.registration_dat...	15 row(s) returned	0.062 sec / 0.000 sec

Figure 4.6 — Q4.6 result: dormant customers]

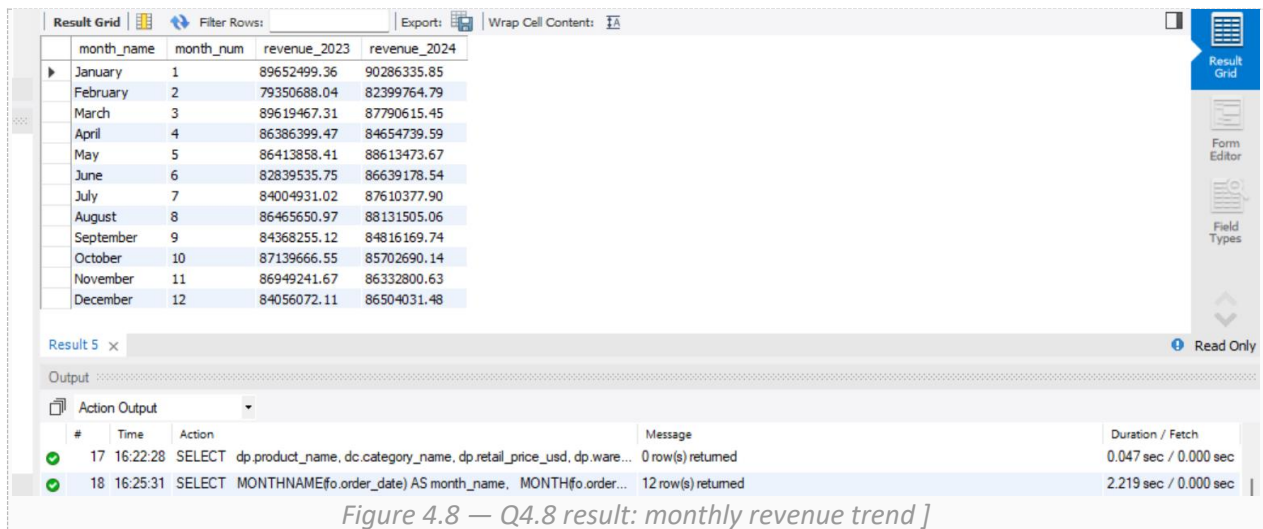
Q4.7 — Dead Stock & Capital Audit

Identifies products with warehouse stock that have never been ordered, and calculates the capital tied up in that unsold inventory (stock × unit cost).



Q4.8 — Monthly Revenue Trend (2023 vs 2024)

Compares net revenue month-by-month across the two years of data available, to surface seasonal or growth trends.



Phase 5: Transactions & Data Integrity

This phase demonstrates how MySQL transactions protect a data warehouse when an update goes wrong midway through a batch of changes. A transaction was opened, a correct change was made and checkpointed, a deliberate error was introduced and undone, and the remaining correct change was committed — all without losing the first correct change.

5.1 Cost Adjustment for Renewable Energy & Solar

Inside an open transaction (START TRANSACTION), the unit cost of every product in the Renewable Energy & Solar category was increased by 15% to reflect updated supplier pricing. A savepoint named after_solar_update was then created to mark this correct, validated state.

5.2 Deliberate Error and Proof of Corruption

To simulate a real-world mistake — a junior analyst running an unfiltered UPDATE — every product's unit_cost_usd was overwritten to a flat 50.00, regardless of category. A SELECT query confirmed the corruption across the table.

5.3 Partial Rollback

ROLLBACK TO SAVEPOINT after_solar_update reversed only the erroneous flat-cost update. Because the savepoint was created after the Solar adjustment, that earlier, correct 15% increase was preserved. This is the core lesson of the exercise: savepoints allow surgical recovery from an error without discarding valid work completed earlier in the same transaction.

5.4 Correct EV Mobility Adjustment and Commit

With the transaction restored to its clean state, the unit cost of all EV Mobility & Charging products was increased by 8% to reflect a separate, correct pricing update. COMMIT then made both the Solar and EV Mobility changes permanent.

[INSERT SCREENSHOT HERE — Figure 5.1 — Output panel showing ROLLBACK TO SAVEPOINT, the 40-row EV UPDATE, and COMMIT, all with green success indicators]

Figure 5.1: Transaction log showing rollback to savepoint, EV update (40 rows affected), and commit.

5.5 Verification

After the commit, a validation query confirmed that no products remained at the erroneous 50.00 cost. A second query compared each affected product's new cost to its original raw cost, finding an average multiplier of 1.15 for the 40 Renewable Energy & Solar products and 1.08 for the 40 EV Mobility & Charging products — confirming each adjustment was applied exactly once, correctly.

[INSERT SCREENSHOT HERE — Figure 5.2 — Verification query result showing avg_multiplier of 1.15 (Solar) and 1.08 (EV Mobility)]

Figure 5.2: Verification of cost adjustments (Solar $\times 1.15$, EV Mobility $\times 1.08$).

Phase 6: Views & Reporting Layer

Three views were built on top of the production schema to give business users repeatable, self-service access to key metrics without needing to write SQL themselves.

6.1 Monthly Revenue Dashboard (vw_monthly_revenue_dashboard)

Summarizes units sold, net revenue, and gross profit by year, month, and product category. Returns 120 rows (24 months across 5 categories).

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

revenue_year	revenue_month	month_name	category_name	units_sold	net_revenue	gross_profit
2023	1	January	Eco-Friendly Home & Climate	13760	8667693.31	2962169.41
2023	1	January	EV Mobility & Charging	14010	33979012.04	10688158.36
2023	1	January	Industrial Energy Efficiency	13846	24853180.45	8752006.76
2023	1	January	Renewable Energy & Solar	14029	11538166.67	2982351.12
2023	1	January	Smart Grid & IoT Energy	13505	10614446.89	3749582.93
2023	2	February	Eco-Friendly Home & Climate	12761	7963774.46	2717845.80
2023	2	February	EV Mobility & Charging	12344	29006490.96	8950573.37
2023	2	February	Industrial Energy Efficiency	12623	21895418.91	7574481.56
2023	2	February	Renewable Energy & Solar	13014	10665356.82	2799510.44
2023	2	February	Smart Grid & IoT Energy	12772	9819646.89	3429534.54
2023	3	March	Eco-Friendly Home & Climate	13552	8704604.51	2945338.24
2023	3	March	EV Mobility & Charging	14371	34966526.54	10771817.25
2023	3	March	Industrial Energy Efficiency	13725	24389381.60	8585070.06
2023	3	March	Renewable Energy & Solar	13566	10899207.49	2796083.59

vw_monthly_revenue_dashboar... x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	12:00:17	SELECT * FROM vw_dead_stock_alert	0 row(s) returned	0.765 sec / 0.000 sec
12	12:06:00	SELECT * FROM vw_monthly_revenue_dashboard ORDER BY revenue_yea...	120 row(s) returned	6.125 sec / 0.000 sec

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

revenue_year	revenue_month	month_name	category_name	units_sold	net_revenue	gross_profit
2023	3	March	Smart Grid & IoT Energy	13859	10659747.17	3719680.23
2023	4	April	Eco-Friendly Home & Climate	13269	8657672.47	3005203.66
2023	4	April	EV Mobility & Charging	13608	32472008.15	9881426.21
2023	4	April	Industrial Energy Efficiency	13459	23981578.01	8482626.34
2023	4	April	Renewable Energy & Solar	13861	11194147.24	2878574.27
2023	4	April	Smart Grid & IoT Energy	13098	10080993.60	3551926.80
2023	5	May	Eco-Friendly Home & Climate	13650	8622811.39	2973901.31
2023	5	May	EV Mobility & Charging	13830	31447312.92	9576282.67
2023	5	May	Industrial Energy Efficiency	13745	24646784.97	8696561.55
2023	5	May	Renewable Energy & Solar	14198	11592494.02	2954607.03
2023	5	May	Smart Grid & IoT Energy	13434	10104455.11	3542536.64
2023	6	June	Eco-Friendly Home & Climate	13377	8437982.52	2849951.68
2023	6	June	EV Mobility & Charging	12955	29902966.35	9191881.29
2023	6	June	Industrial Energy Efficiency	13448	23986712.21	8465573.98

vw_monthly_revenue_dashboar... x

Output

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Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

revenue_year	revenue_month	month_name	category_name	units_sold	net_revenue	gross_profit
2023	6	June	Renewable Energy & Solar	13203	10532074.23	2697022.81
2023	6	June	Smart Grid & IoT Energy	13461	9979800.44	3490736.82
2023	7	July	Eco-Friendly Home & Climate	14006	9086713.60	3089722.17
2023	7	July	EV Mobility & Charging	13641	30495143.84	9380571.67
2023	7	July	Industrial Energy Efficiency	13179	22616095.86	7884341.26
2023	7	July	Renewable Energy & Solar	13902	11040400.36	2818920.98
2023	7	July	Smart Grid & IoT Energy	14092	10766577.36	3738378.89
2023	8	August	Eco-Friendly Home & Climate	13780	9002730.11	3090007.15
2023	8	August	EV Mobility & Charging	13606	30937852.57	9382256.40
2023	8	August	Industrial Energy Efficiency	13844	24563736.66	8707837.87
2023	8	August	Renewable Energy & Solar	14099	11413555.93	2958343.21
2023	8	August	Smart Grid & IoT Energy	13979	10547775.70	3714733.08
2023	9	September	Eco-Friendly Home & Climate	13579	8655225.09	2946694.25
2023	9	September	EV Mobility & Charging	13116	29882751.07	8974789.55

vw_monthly_revenue_dashboar... x

Output

Action Output

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11	12:00:17	SELECT * FROM vw_dead_stock_alert	0 row(s) returned	0.765 sec / 0.000 sec
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Result Grid

Filter Rows:

Export:

Wrap Cell Content:

revenue_year	revenue_month	month_name	category_name	units_sold	net_revenue	gross_profit
2023	9	September	Industrial Energy Efficiency	13492	24165168.91	8583030.86
2023	9	September	Renewable Energy & Solar	13947	11207221.18	2839955.57
2023	9	September	Smart Grid & IoT Energy	13752	10457888.87	3647583.54
2023	10	October	Eco-Friendly Home & Climate	13744	8694927.79	3009549.62
2023	10	October	EV Mobility & Charging	13479	30616341.18	9252600.90
2023	10	October	Industrial Energy Efficiency	14415	26282329.31	9269273.18
2023	10	October	Renewable Energy & Solar	13560	11013560.82	2823621.56
2023	10	October	Smart Grid & IoT Energy	13932	10532507.45	3710100.16
2023	11	November	Eco-Friendly Home & Climate	13262	8521166.33	2892217.53
2023	11	November	EV Mobility & Charging	13418	32846664.35	10188173.41
2023	11	November	Industrial Energy Efficiency	13552	24399287.74	8599202.53
2023	11	November	Renewable Energy & Solar	13569	11199044.41	2877653.24
2023	11	November	Smart Grid & IoT Energy	13527	9983078.84	3501802.01
2023	12	December	Eco-Friendly Home & Climate	13735	8692287.60	2957769.53

vw_monthly_revenue_dashboar... x

Read Only

Output

Action Output

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Result Grid

Filter Rows:

Export:

Wrap Cell Content:

revenue_year	revenue_month	month_name	category_name	units_sold	net_revenue	gross_profit
2023	12	December	EV Mobility & Charging	13792	30531110.31	9405007.04
2023	12	December	Industrial Energy Efficiency	12991	23081802.38	8076329.71
2023	12	December	Renewable Energy & Solar	13840	11205674.77	2912933.16
2023	12	December	Smart Grid & IoT Energy	13838	10545197.05	3664739.05
2024	1	January	Eco-Friendly Home & Climate	13658	8847587.59	3035732.08
2024	1	January	EV Mobility & Charging	13908	34503901.24	10582496.14
2024	1	January	Industrial Energy Efficiency	14050	25173703.97	8835800.13
2024	1	January	Renewable Energy & Solar	13487	10873303.11	2815468.65
2024	1	January	Smart Grid & IoT Energy	13936	10887839.94	3786630.58
2024	2	February	Eco-Friendly Home & Climate	12823	8140093.89	2797260.23
2024	2	February	EV Mobility & Charging	13585	31674581.49	9675045.87
2024	2	February	Industrial Energy Efficiency	12762	22637002.95	7872796.61
2024	2	February	Renewable Energy & Solar	12692	10134068.01	2581062.77
2024	2	February	Smart Grid & IoT Energy	12842	9814018.45	3396833.07

vw_monthly_revenue_dashboar... x

Read Only

Output

Action Output

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11	12:00:17	SELECT * FROM vw_dead_stock_alert	0 row(s) returned	0.765 sec / 0.000 sec
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Result Grid

Filter Rows:

Export:

Wrap Cell Content:

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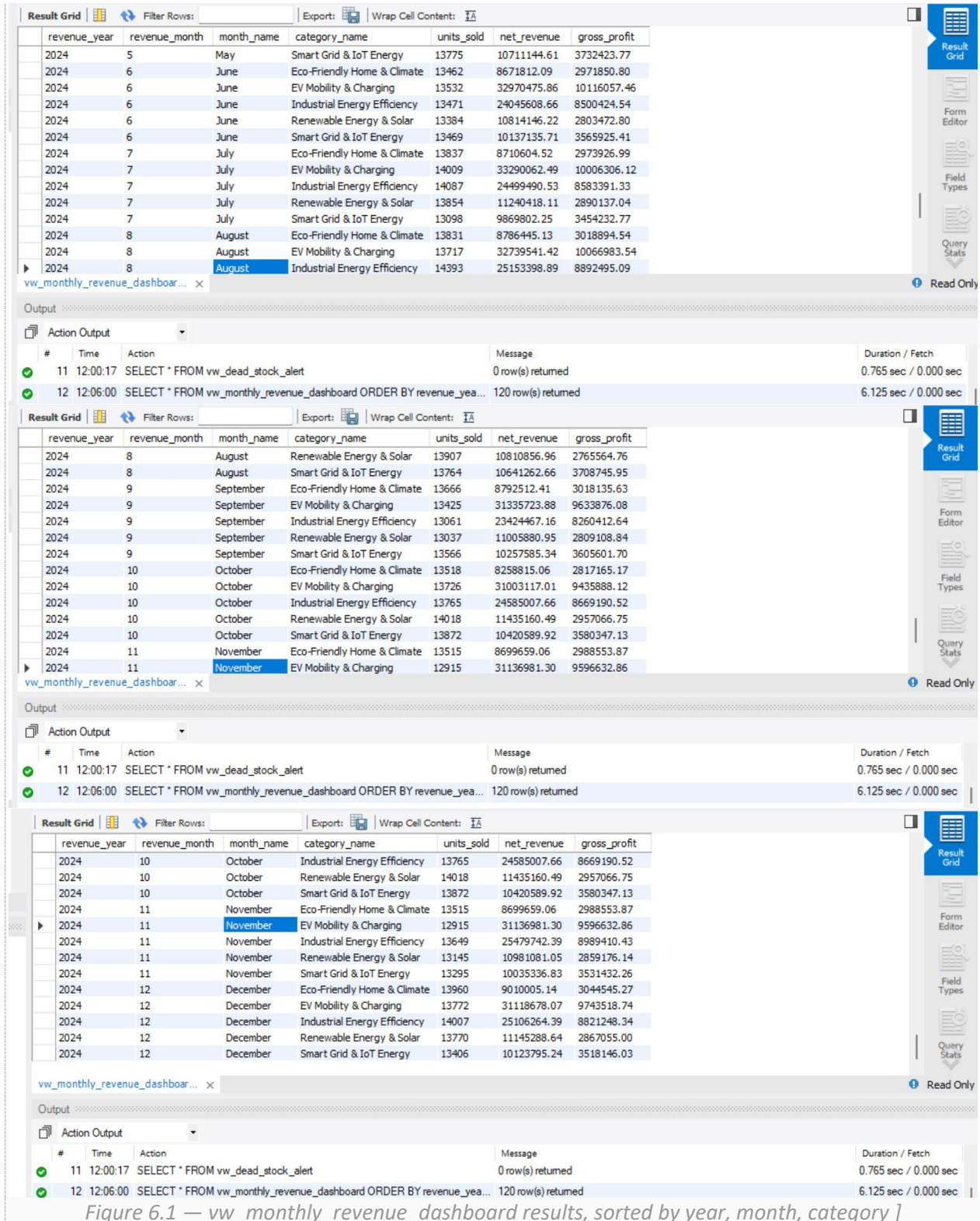


Figure 6.1: Monthly revenue dashboard results.

6.2 Customer 360 (vw_customer_360)

Provides a single view of every customer: lifetime order count, total spend, average order value, last order date, and a loyalty tier (Elite / Preferred / Standard / New) derived from the customer's stored loyalty score. A LEFT JOIN ensures all 15,000 customers appear, including 800 who have never placed an order, which supports dormant-customer analysis. Because the loyalty tier is derived from the stored score rather than from purchase behavior, some customers with zero orders can still hold a high tier — a known characteristic of the source data, noted here for transparency.

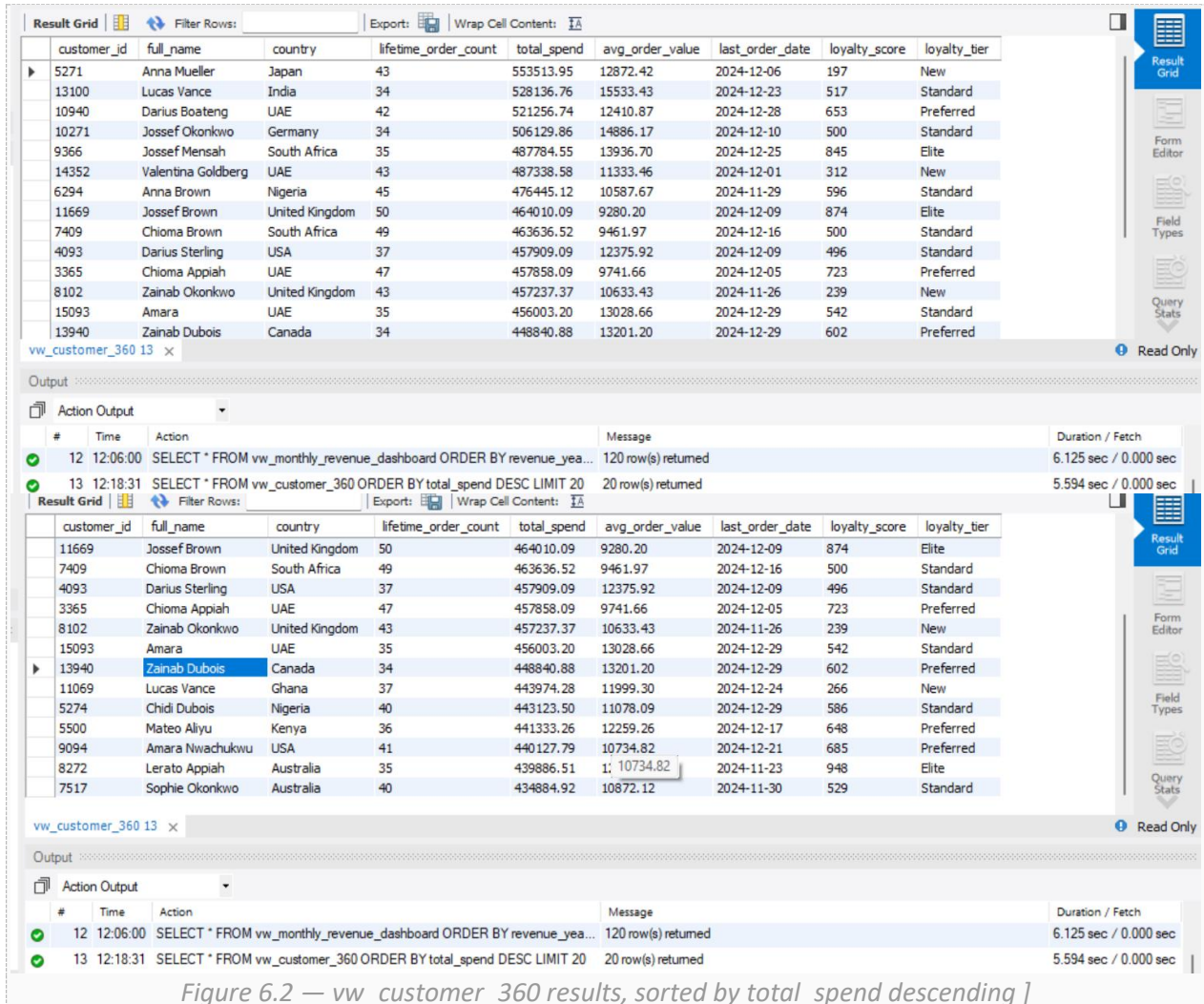


Figure 6.2 — vw_customer_360 results, sorted by total_spend descending]

Figure 6.2: Customer 360 view results.

6.3 Dead Stock Alert (vw_dead_stock_alert)

Flags products that have stock on hand (warehouse_stock > 0) with no orders in the 90 days before the latest order date in the dataset (31 December 2024), and calculates the capital tied up in that inventory (stock × unit cost). The original version of this view only flagged products that had never sold a single unit, which could not detect genuinely slow-moving stock; the logic was corrected to use a rolling 90-day sales window instead.

Validation confirmed that all 200 products in the catalog recorded at least one sale within the final 90 days of the dataset, so the view correctly returns zero rows for this dataset — there is no dead stock to

report given the current sales activity. This is a valid business result, not a defect in the view, and was confirmed by cross-checking staging-to-production row counts (200 raw products = 200 clean products; 500,000 raw orders = 500,000 clean orders), which ruled out data loss as an explanation for the empty result.

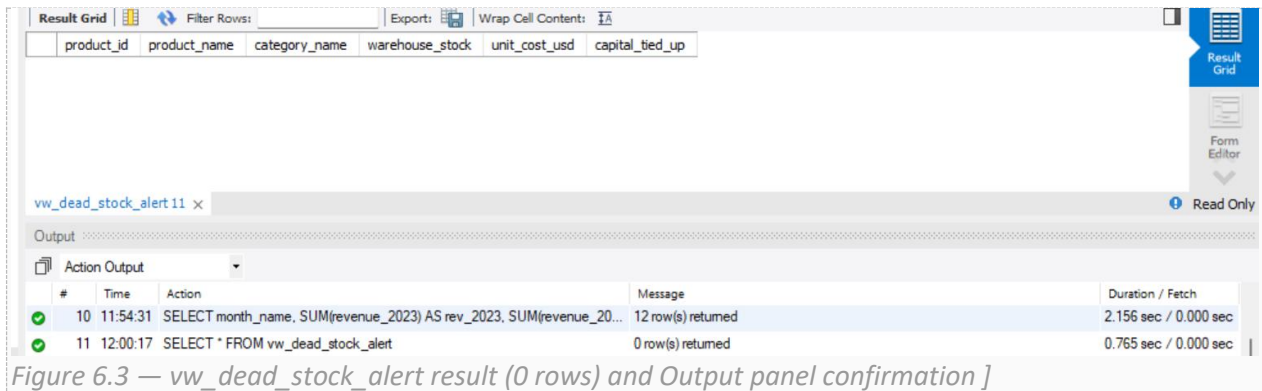


Figure 6.3 — *vw_dead_stock_alert* result (0 rows) and Output panel confirmation]

Figure 6.3: Dead stock alert view — zero rows, confirming full product turnover in the dataset.

Conclusion

The TerraPulse data warehouse successfully consolidates five heterogeneous data sources into a single, validated, multi-currency analytical platform. The ETL pipeline resolved inconsistent date formats, currency codes, casing, and missing values with zero record loss between staging and production. Eight business-intelligence queries answer TerraPulse's core reporting questions around profitability, geography, currency risk, logistics performance, and customer value. A transaction-safety exercise demonstrated recovery from a simulated data-entry error using savepoints without losing valid, already-applied work. Finally, three production views give business users repeatable, self-service access to monthly performance, a full customer profile, and inventory health — completing the warehouse's role as a durable reporting layer for the organization.

Known Limitations

- Customer loyalty tier is derived from a stored score rather than actual purchase behavior, so a small number of high-tier customers have no order history.
- Dead-stock detection currently uses a fixed 90-day window measured from the latest order date in the dataset; this should be revisited against a live, rolling 'today' once the warehouse is in production.